Google Cloud H2S

AI Builder Cup

Build in JAPAC.
Build for the world.



Team Details

a. Team name: **Canon**
b. Team leader name: **Tejas Nafde**
c. Problem Statement: **Future of Work & Enterprise Productivity**

Brief about the idea

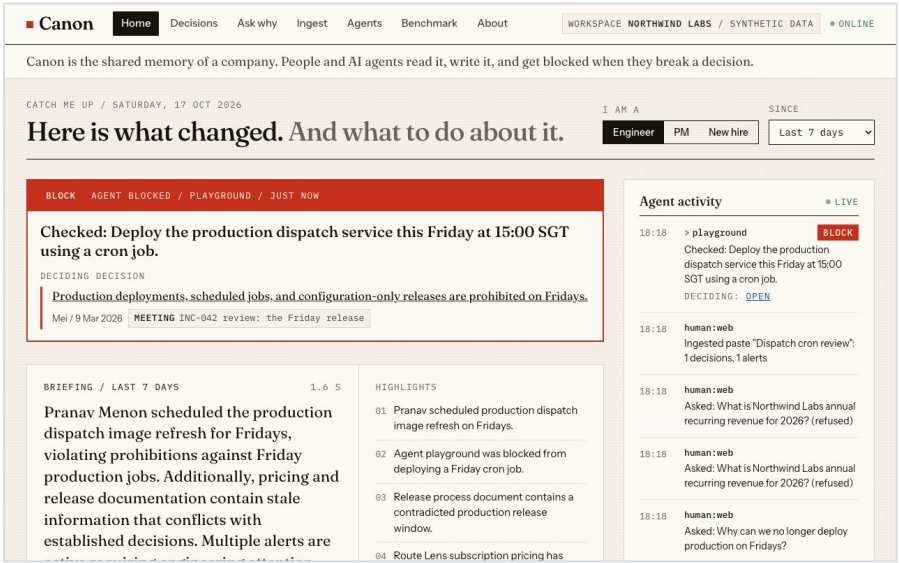
Canon is the shared memory of a company, for its people and for its AI agents.

Teams decide things in Slack threads, meetings, PRs and email. The wiki goes stale. New hires and AI agents act on old information.

Canon keeps one record of decisions. Each decision has a rationale, an owner, a date and a quote checked word for word against its source.

Every new message, document and agent action is checked against that record. Canon flags drift: a contradiction, a stale wiki page, a broken premise.

Coding agents connect over MCP and call check_intent before they act. A Friday cron deploy is blocked, with the incident review as evidence.



Opportunities

How is it different?	How does it solve the problem?	USP: agent-native
- Meeting-memory tools (Mnemo style) summarise. Decision logs (Decision Ledger style) store. Spec checkers (Spec Keeper style) review code. - Canon does all three on one record, and its first-class users include AI agents, not only people.	- Extracts decisions with verified quotes from any source. - Reconciles each one: same, update or contradiction. - Raises alerts on stale docs and broken premises. - Answers why, with citations, or says I do not know.	- An MCP server any agent can call: check_intent returns allow, warn or block with the deciding decision. - Every agent read, write and check shows in a live feed. - Also an ADK agent over A2A.

List of features offered by the solution

Ingest any source: Slack, meeting, wiki, PR, commit, email, free paste, image or PDF.

Decision extraction: statement, rationale, assumptions, owner, date, verified quotes.

Reconciliation: same, update (supersedes) or contradiction.

Drift alerts: stale wiki pages and premise changes, both quotes side by side.

Ask why: cited answer and decision timeline; refuses when the record is silent.

Catch me up: engineer, PM or new-hire briefing with ranked next actions (score 0 to 100 with reasons) and who to ask.

Agent guard: check_intent gives allow, warn or block with the deciding decisions.

MCP server: six tools over streamable HTTP for Claude Code and Codex.

Live agent activity feed and an agent playground in the browser.

Ask the departed expert: answers in a former colleague's voice, from their quotes only.

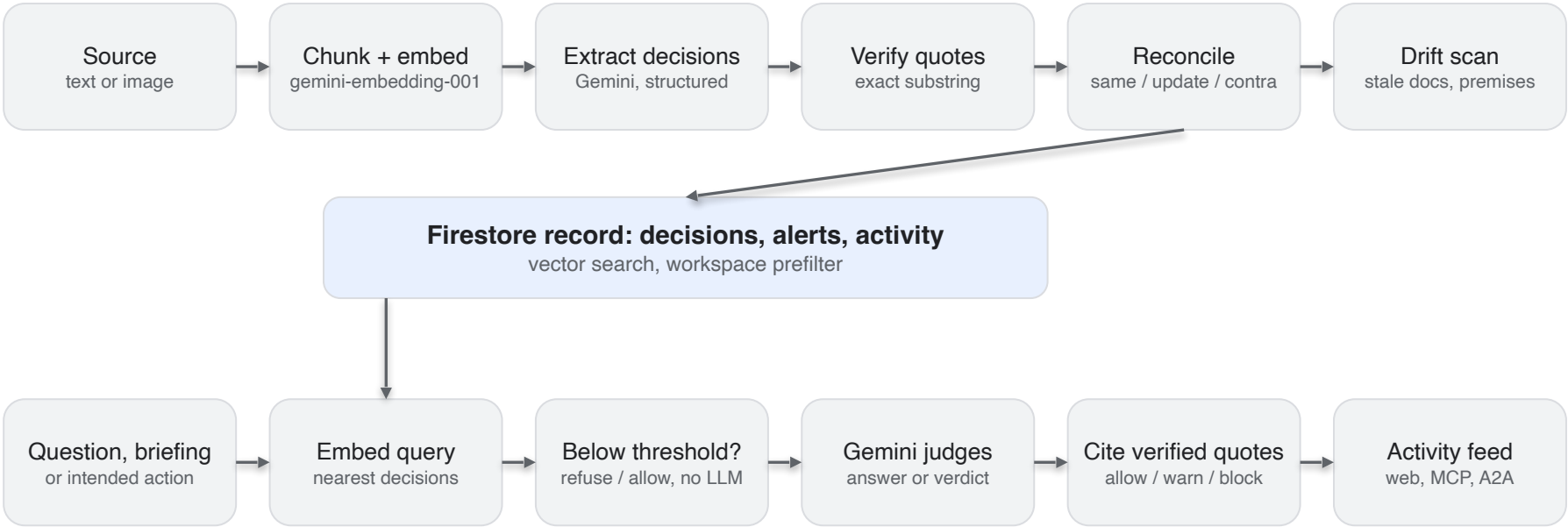
Human in the loop: alerts stay open until a person resolves them.

Integrations: A2A agent card, GitHub PR webhook, Discord slash command.

Benchmark: planted problems vs found, on a labelled gold set.

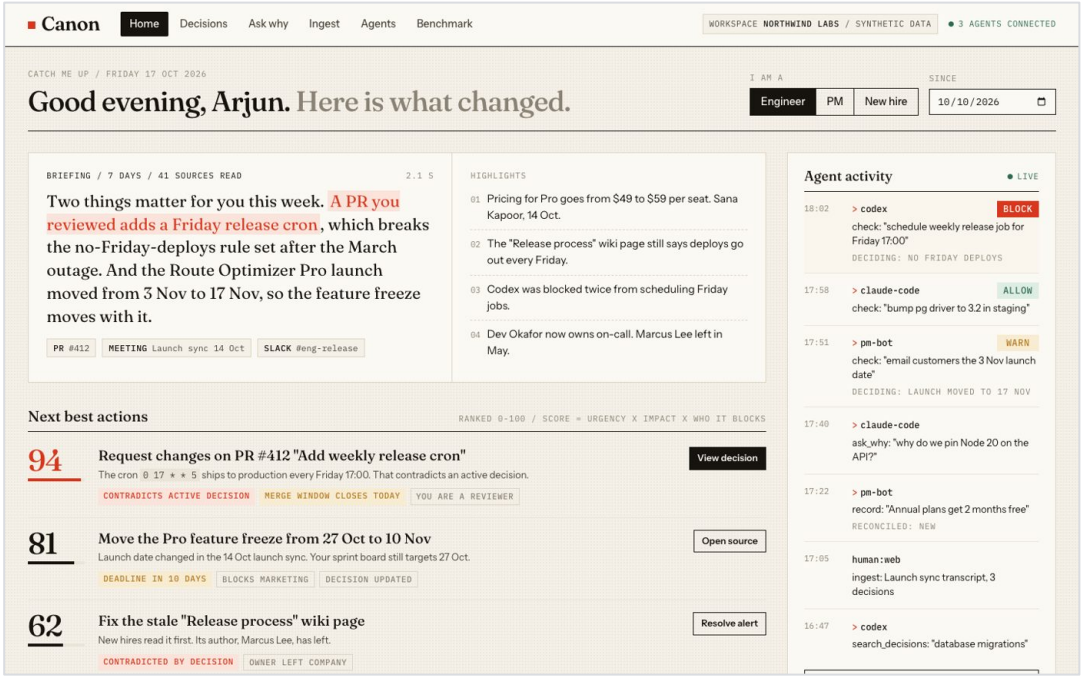
Process flow diagram

A. A new source arrives (web paste, Slack, PR, agent)

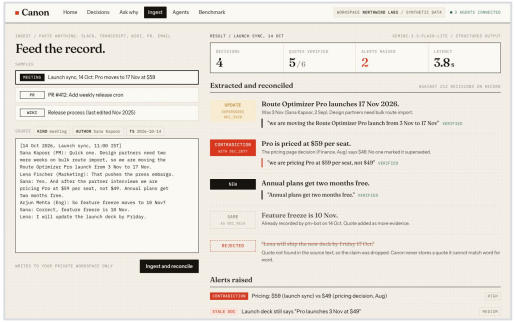


B. A person or an agent asks or acts

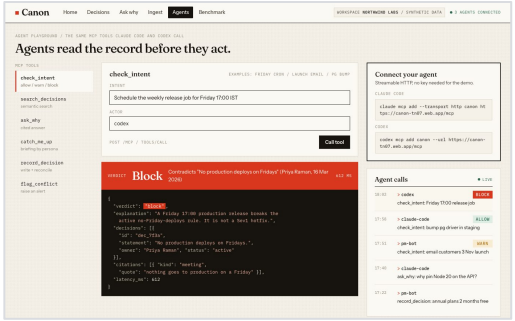
Wireframes / mock diagrams



Home: briefing, alerts, agent feed

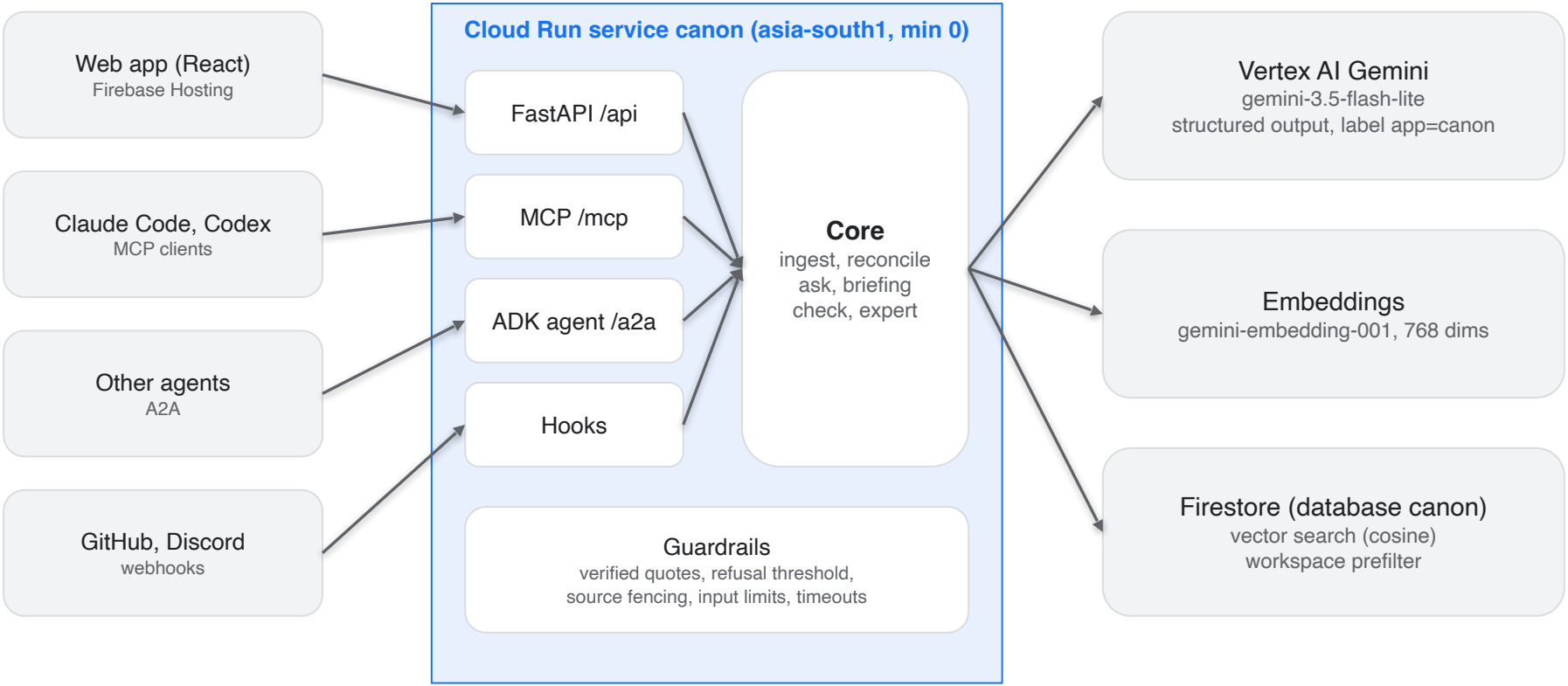


Ingest: paste and reconcile



Agents: MCP playground

Architecture diagram



Technologies used in the solution

Technology	What Canon uses it for
Gemini 3.5 Flash-Lite on Vertex AI	Extraction, reconciliation, drift checks, answers, briefings, check_intent verdicts; structured output (response_schema)
gemini-embedding-001 (768 dims)	Embeddings for source chunks, decisions and queries
Cloud Firestore, named database canon	Sources, chunks, decisions, alerts, activity, people; vector search with a workspace prefilter
Cloud Run	One scale-to-zero service: FastAPI API, MCP server, A2A agent, GitHub and Discord hooks
Firebase Hosting	Static web app; rewrites /api, /mcp, /a2a to Cloud Run on one origin
Agent Development Kit (ADK) + A2A	Canon as an LlmAgent with four tools, discoverable by agent card
Model Context Protocol (FastMCP)	Six tools over streamable HTTP for Claude Code, Codex and other clients
Python 3.12, FastAPI, uv, pytest	Backend; tests replay recorded Gemini fixtures, no network
React, TypeScript, Vite	Web app: home, decisions, ask, ingest, agents, benchmark

Estimated implementation cost

Idle cost is near zero. Cloud Run scales to zero (min instances 0). There is no per-hour resource.

Firestore: the demo data fits inside the free tier so far.

Gemini is paid per call. An answer is one embedding and one Flash-Lite call. A check is one embedding and one call, or none when no decision is relevant. An ingest is one extraction call plus a few reconcile calls.

Refusals and irrelevant checks never reach Gemini. The similarity threshold stops them first.

Tracked, not guessed: every Gemini call and the Cloud Run service carry the label app=canon, so the billing export splits Canon's spend.

What we measured (2026-10-17):

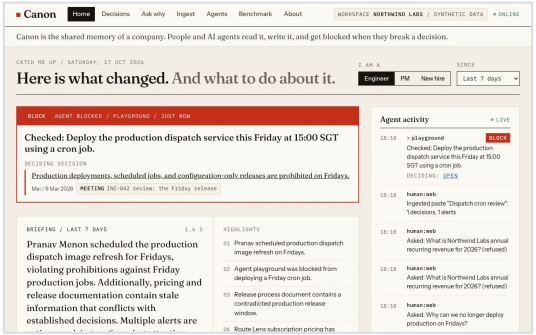
- The billing export has no app=canon rows yet. It lags by up to a day.

- All embedding usage on the shared billing project since 2026-10-12: INR 0.16.

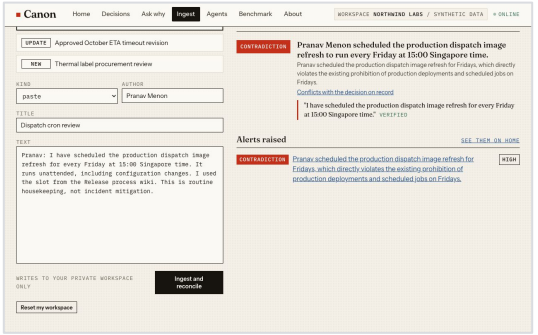
List price (unverified): third-party listings quote USD 0.30 per 1M input tokens and USD 2.50 per 1M output tokens for Gemini 3.5 Flash-Lite. Check the Vertex AI pricing page before you rely on it.

Not measured yet: tokens per call, so we do not quote a per-call price.

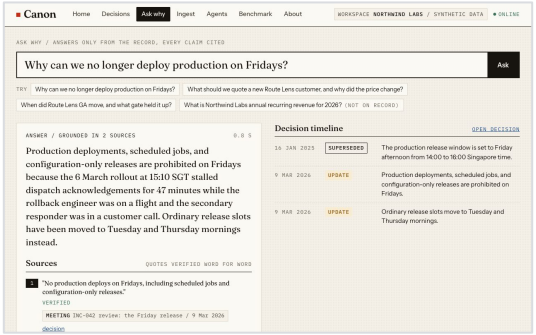
Snapshots of the prototype



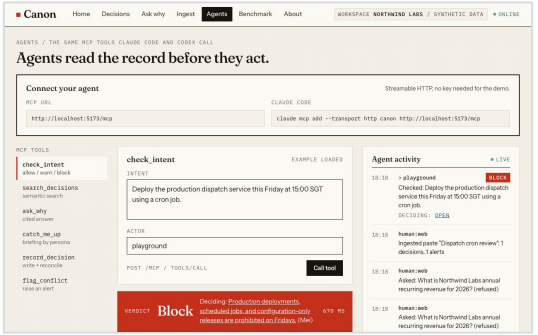
Agent blocked on a Friday deploy



Paste raises a contradiction



Ask why: cited answer and timeline



Agent playground over MCP

Prototype performance report / benchmarking

Task (labelled gold set)	Planted	Found	Score
Decisions extracted	38	37	97%
Contradictions and updates	17	13	76%
False positive alerts		2	
check_intent verdicts	15	13	87%
Unanswerable, refused	5	5	100%
Latency p50 / p95		1079 / 1325 ms	

Run 2026-10-17 12:47 UTC, model gemini-3.5-flash-lite. Corpus: 259 synthetic sources, about 100 of them pure noise.

Reproduce: backend/scripts/seed.py, then backend/scripts/eval.py. The app serves the report at /api/eval.

Notes and misses, in the open

- stale docs found 7/8
- premise changes found 4/5
- updates found 7/9, contradictions found 6/8
- answerable questions answered 10/10, citing an expected decision 10
- 206 stored quotes, every one verified word for word against its source
- latency over 30 ask and check calls, sequential, client to Vertex AI and Firestore in asia-south1
- check miss check-05: expected allow, got block
- check miss check-09: expected block, got allow
- live paste samples: relation correct 3/3, ingest p50 2249 ms

Additional details / future development

Where people already talk: Slack, Microsoft Teams and Discord plug-ins that ingest threads and answer in place. A Discord slash command exists today.

Permissions: sign-in, per-team visibility, and decisions that follow the access rules of their source.

Code review at scale: the GitHub webhook already checks PR text; next, check diffs and post review comments for every repo.

Beyond software: manufacturing SOP drift. A line change or a supplier note that contradicts the approved standard operating procedure raises an alert before a shift runs it.

Better recall: the benchmark shows 4 of 17 contradictions missed; we will tune reconciliation against the gold set.

Real data: ingest this repository's own README, PRs and commits next to the synthetic corpus.

Links

GitHub public repository: https://github.com/tejasnafde/canon

Demo video: (add the video link)

Final product link: https://canon-tn07.web.app

Cloud Run API: https://canon-ezvux7gqxa-el.a.run.app

Connect your agent: claude mcp add --transport http canon https://canon.tn07.dev/mcp

No login needed. All company data is synthetic and labelled as such.

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Thank you

